

# Series and infinite products – 1

Convergence: the standard scale and comparison

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## Theoretical minimum.

**Theorem 1 (comparison and the standard scale).** For series with non-negative terms,  $a_n \leq b_n$  gives one implication, and  $a_n \sim b_n$  gives equivalence of convergence. Everything is compared with the scale

$$\sum \frac{1}{n^p}, \quad \sum \frac{1}{n \ln^p n}, \quad \sum \frac{1}{n \ln n \ln^p \ln n}, \quad \dots$$

each of which converges exactly for  $p > 1$ ; no member of this scale is a boundary for the previous one, so there is no slowest convergent series.

**Theorem 2 (Cauchy condensation).** If  $a_n \geq 0$  is non-increasing, then  $\sum a_n$  and  $\sum 2^k a_{2^k}$  converge or diverge together. This is the cheapest way to walk along the whole scale above, and the monotonicity may not be dropped.

**Theorem 3 (integral test).** If  $f$  is non-negative and non-increasing on  $[1, \infty)$ , then  $\sum f(n)$  and  $\int_1^\infty f$  converge together, and  $0 \leq \sum_{k \leq n} f(k) - \int_1^n f \leq f(1)$ ; the difference always has a limit, which for  $f(x) = 1/x$  is Euler's constant.

**Theorem 4 (Raabe and Gauss).** When d'Alembert's ratio tends to 1, expand it: if  $\frac{a_{n+1}}{a_n} = 1 - \frac{p}{n} + O\left(\frac{1}{n^2}\right)$ , then the series converges for  $p > 1$  and diverges for  $p < 1$ . This is exactly the sensitivity the ratio test lacks; the logarithmic refinement of Bertrand settles  $p = 1$ .

**Fact 1 (where comparison breaks).** The equivalence  $a_n \sim b_n$  transfers convergence only for eventually one-signed terms: for  $a_n = \frac{(-1)^n}{\sqrt{n}}$  and  $b_n = \frac{(-1)^n}{\sqrt{n}} + \frac{1}{n}$  one has  $a_n \sim b_n$ , yet one series converges and the other does not. For sign-changing series the tools are in Series and infinite products – 3.

1. Prove the condensation theorem and the integral test, and use them to walk along the whole scale: determine for which  $p$  the series  $\sum \frac{1}{n \ln^p n}$  and  $\sum \frac{1}{n \ln n \ln^p \ln n}$  converge. Deduce that for every convergent series with positive terms there is a convergent series with larger terms, and for every divergent one a divergent series with smaller terms.
2. (IMC 2018) Let  $(a_n)$  and  $(b_n)$  be two sequences of positive numbers. Show that the following statements are equivalent:
  - 1) there is a sequence  $(c_n)$  of positive numbers such that  $\sum \frac{a_n}{c_n}$  and  $\sum \frac{c_n}{b_n}$  both converge;
  - 2)  $\sum \sqrt{\frac{a_n}{b_n}}$  converges.
3. (IMC 2019) Let  $C = \{4, 6, 8, 9, 10, \dots\}$  be the set of composite positive integers. For each  $n \in C$  let  $a_n$  be the smallest positive integer  $k$  such that  $k!$  is divisible by  $n$ . Determine whether the series  $\sum_{n \in C} \left(\frac{a_n}{n}\right)^n$  converges.
4. (VJIMC 2004) Let  $\sum a_n$  be a divergent series with positive non-increasing terms. Prove that the series  $\sum \frac{a_n}{1+na_n}$  diverges.
5. (SEEMOUS 2024) Let  $(x_n)_{n \geq 1}$  be defined by  $x_1 \in (0, 1)$  and  $x_{n+1} = x_n - \frac{x_n^2}{\sqrt{n}}$  for all  $n \geq 1$ . Find the values of  $\alpha \in \mathbb{R}$  for which the series  $\sum x_n^\alpha$  converges.